XIAMEN UNIVERSITY MALAYSIA
厦門大學馬來西亞分校

EEE312 Week 11
Lab Practice

Evaluate the following integrals on the given data using the trapezoidal method.

| $x$ | 0 | 0.25 | 0.5 | 0.75 | 1.0 | 1.25 | 1.5 |
|---|---|---|---|---|---|---|---|
| $y$ | −1 | 1 | 2.5 | 4 | 2 | 1.5 | 1 |

// Beginning of the C Source code

**Function main()**

```
1   // Integration using trapezoidal rule
2   #include <stdio.h>
3
4   int main()
5   {
6       float x[7] = {0, 0.25, 0.5, 0.75, 1.0, 1.25, 1.5};
7       float y[7] = {-1, 1, 2.5, 4, 2, 1.5, 1};
8       float h = x[1] - x[0];
9       float sum = 0.0; // sum the y values from i=1 to i=5.
10      float integration = 0.0; // integration result
```

Line 6 to 7: Declare and initialize x and y as given in the question.

Line 8: Declare a variable h which is the distance between two consecutive x values.

Line 9: Declare a variable sum to store the sum of y[1] to y[5].

Line 10: Declare a variable integration to store the result of the integration.

# Trapezoidal Rule Implementation

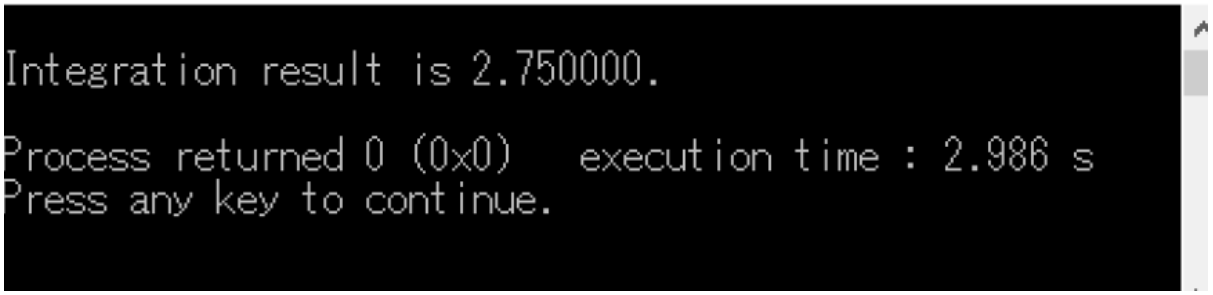
```
12      for(int i = 1; i<6; i++)
13          sum += y[i];
14
15      integration = (h/2.0)*(y[0]+y[6]+2.0*sum);
16
17      printf("\nIntegration result is %f.\n", integration);
18
19      return 0;
20  }
```

Line 12-13: Sum all y[i]except for y[0]and y[6].

Line 15: Implement the trapezoidal rule equation in C code:

$$\int_{x_0}^{x_6} f(x)dx \approx \frac{h}{2}[y_0 + y_6 + 2(y_1 + y_2 + y_3 + y_4 + y_5)]$$

## Output


Integration result is 2.750000.
Process returned 0 (0x0)   execution time : 2.986 s
Press any key to continue.